

Complex-Valued Phase-Coherent Transformer

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Abstract

Complex-valued Transformers have largely inherited softmax attention from real-valued architectures. However, row-normalised token competition is not necessarily aligned with phase-preserving computation. In this paper, we introduce the Phase-Coherent Transformer (PCT), which applies a real-valued, element-independent, smooth gate to L2-normalised complex query-key similarities. PCT replaces token competition with token-non-competing attention and is designed to preserve phase information across layers.

Across mid-scale benchmarks spanning long-range memory, hierarchical long-range reasoning, positional retrieval, phase-based memory and superposition, and image classification, PCT shows strong generalisation across task categories. Under parameter-fair comparison, PCT consistently outperforms both the standard softmax Transformer and its direct complex-valued counterpart. Moreover, even on tasks traditionally considered difficult for complex-valued neural networks, such as NIAH and LRA-Text, PCT remains competitive with Multiscreening, the strongest real-valued NN baseline in our comparison. Experiments introducing gates that deliberately violate the PCT conditions show that the design is not incidental: smooth gates that preserve negatively aligned phase components remain strong, whereas gates that delete such components collapse on long-range retrieval, and gates whose outputs become excessively large suffer clear performance degradation. PCT also shows no depth-related accuracy collapse across the tested depth range. These results support introducing multi-layer phase-coherent structure into attention as a promising design principle for achieving generalisation in complex-valued Transformers.

1 Introduction

Complex-valued neural networks (CvNNs) have long not been viewed as candidates for general-purpose architectures. The literature that has developed since Trabelsi 2017 has centred on signal-processing domains where the input is intrinsically represented as complex numbers – PolSAR classification, wireless communication, SAR target recognition, MRI, and audio spectrogram analysis. The implicit understanding has been that CvNNs are strong on complex-valued signals but cannot match real-valued NNs on general-purpose domains such as text, vision, and symbolic reasoning.

Behind this lies an underdevelopment of the architectural research itself. While complex extensions of convolutions, normalisation layers, and activations have been studied systematically, the Transformer side – including well-designed enhancements oriented towards image tasks such as Eilers CAtt and Holographic Transformer – has not seriously explored anything beyond softmax attention. Attention mechanisms genuinely fitted to a complex-valued substrate – primitives that depart substantially from softmax, which was optimised for real-valued NNs – have not been systematically investigated.

Looking at the situation from a different angle, two observations point in the same direction:

1. **On the real-valued NN side**, Multiscreening (Nakanishi 2026) has shown that an attention primitive can substantially outperform softmax Transformers on long-range dependency tasks, undermin-

ing the assumption that softmax attention is universally optimal. Room to fundamentally redesign the attention primitive has opened up even for real-valued NNs.

2. **On the complex-valued NN side specifically**, complex weights and inputs carry phase information, and a deep network can perform meaningful computation only if phase information is preserved across layers. There is no a priori reason to expect that attention structures developed for real-valued NNs – and the row normalisation of softmax in particular – would contribute to this multi-layer phase preservation. On the contrary, softmax row normalisation structurally introduces a "token competition" in which tokens compete for attention mass, and this acts as an unnecessary constraint on a complex computation whose layers must preserve phase.

Combining these two observations, a natural conclusion follows: what should be tried first in a complex NN is **token non-competition** – an attention primitive whose weights are not normalised to sum to one within a row. This paper formalises the idea as the **Phase-Coherent Transformer (PCT)**: a sigmoid gate on the cosine score of L2-normalised complex Q, K used as the attention weight, with no row normalisation whatsoever. It is the most natural lift of sigmoid attention (Ramapuram 2024) to complex space, and forms a single family together with neighbouring variants such as tanh+1, softplus, and the complex version of screening attention (Nakanishi 2026).

We carry out a systematic 6-cell comparison on 9 mid-scale benchmarks and show that the PCT family consistently outperforms both existing real- and complex-valued Transformers under parameter-fair conditions ($realdimension = complexdimension \times \sqrt{2}$), achieves 1.000 on long-range NIAH – a task on which CvNNs have traditionally been considered weak – and exhibits no depth collapse over the depth 2–20 scaling range, addressing a long-standing concern about depth scalability in CvNNs.

Results:

1. PCT and its neighbouring cells generalise across multiple tasks (elaborated as H1 in §4).
2. PCT retains dominance on NIAH, where complex Q, K offers no inductive advantage by construction (H2).
3. PCT is uniquely LR/batch-robust on long-range Copy (H3).
4. PCT outperforms the vanilla complex transformer (H4), the vanilla real transformer (H5), and the strongest real-side non-softmax baseline **real_screen** (H6) under parameter-fair conditions.
5. A mathematical interpretation – token non-competition & multi-layer phase preservation – is given in §5, with full proofs in Appendix M.

In short, we experimentally observe strong signs of generalisation in complex-valued neural networks through the introduction of token-non-competing attention and multi-layer phase preservation, and analyzed the conditions of the signs with experiments described in §5.

2 Related Work

2.1 Complex-valued neural networks: Trabelsi 2017; Eilers & Jiang 2023; Holographic Transformer 2024; torchcvnn 2026. Application track: PolSAR classification, wireless modulation, SAR target recognition.

2.2 Real-side attention primitives: softmax (Vaswani 2017); linear / Performer (Choromanski 2020); ReLU attention (Wortsman 2023); sigmoid attention (Ramapuram 2024); screening attention (Nakanishi 2026); polynomial attention (Banerjee 2024). Each of these is a candidate for lifting to complex space.

2.3 Existing complex attention: Eilers CAtt = softmax of $|\langle q, k \rangle|$; APAtt = phase-aware variant with explicit phase term; Holographic Transformer = phase-coherent value rotation; lucidrains' implementations – all use softmax-style attention.

2.4 What is missing: a systematic non-softmax $\times$ complex study, with parameter-fair real/complex comparison, under modern training protocols. This paper fills that gap.

3 Approach

This section presents the 6-cell comparison setup. We first state the two empirical predictions that the PCT hypothesis implies (§3.1), then describe the cell architectures, substrate choices, tasks, training protocol, and hardware.

3.1 Empirical predictions from the PCT hypothesis

The PCT hypothesis – that PCT delivers a special form of generalisation in complex-valued neural networks via **token non-competition** + **multi-layer phase preservation** – implies two specific empirical predictions that the 6-cell comparison directly tests:

Prediction (1) – the "complex penalty" of vanilla complex. On tasks where complex Q, K offers no inductive advantage, the vanilla real transformer outperforms the vanilla complex transformer. Naive softmax-of-complex does not extract useful complex computation; worse, its row normalisation imposes a coupling that is incompatible with the phase superposition of the complex value path. Vanilla complex transformers in the literature inherit this penalty.

Prediction (2) – PCT delivers complex computation. PCT outperforms the vanilla real transformer overall, under parameter-fair conditions ($real_{dim} = complex_{dim} \times \sqrt{2}$). This is the core claim that the PCT architecture extracts useful complex computation that vanilla complex transformers cannot.

These two predictions form the core a priori expectations from the PCT hypothesis. The 6-cell comparison is designed to test them directly:

- `complex_softmax` is the prediction-(1) failing baseline.
- `real_softmax` is the prediction-(2) reference.
- `real_sigmoid`, `real_screen` are additional real-side baselines.
- `complex_screen` is the close-to-PCT cell, sharing PCT’s gate skeleton with a partial deviation.

(Beyond these two predicted observations, two additional findings emerge from the same comparison and are reported in §4.1 alongside the confirmation of (1) and (2).)

3.2 The 6-cell comparison

We compare six attention cells along a 2×3 design:

Cell	Substrate	Gate
<code>real_softmax</code>	real	scaled-dot-product softmax
<code>real_sigmoid</code>	real	element-wise sigmoid, Ramapuram 2024
<code>real_screen</code>	real	screening: $r^2 \cdot \max(s - t, 0)^2$ + post-aggregation TanhNorm + modReLU, Nakanishi 2026
<code>complex_softmax</code>	complex	Eilers CAtt: softmax of $ \langle q, k \rangle $
<code>complex_sigmoid</code> = PCT	complex	sigmoid of $\text{Re}\langle \bar{q}, k \rangle$ on L2-normalised Q, K
<code>complex_screen</code>	complex	screening on $\text{Re}\langle \bar{q}, k \rangle$

Two complex cells are central: **PCT** is the architecture proposed in this paper, with a smooth real-valued element-independent gate on the L2-normalised cosine score; **complex_screen** shares PCT’s gate skeleton but with a hard $(s - t)^2$ threshold, placing it close to PCT in the closeness axis. The remaining four cells serve as parameter-fair baselines for evaluating PCT against vanilla complex / real transformers and against the strongest real-side non-softmax cell.

3.3 Cell architectures

3.3.1 PCT

Given an input token sequence $X \in \mathbb{C}^{N \times d}$, with complex-linear projections $W_q, W_k, W_v, W_o \in \mathbb{C}^{d \times d}$ and a learnable real bias b :

$$\begin{aligned} q_i &= W_q x_i, & k_j &= W_k x_j, & v_j &= W_v x_j \\ \bar{q}_i &= q_i / \|q_i\|_2, & \bar{k}_j &= k_j / \|k_j\|_2 \\ s_{ij} &= \text{Re}\langle \bar{q}_i, \bar{k}_j \rangle \cdot \sqrt{d} \in [-\sqrt{d}, \sqrt{d}] \\ \alpha_{ij} &= \sigma(s_{ij} + b), & b &\text{ initialised to } -\log N \\ \text{out}_i &= W_o \sum_{j=1}^N \alpha_{ij} v_j \end{aligned}$$

Native complex linear for QKV and output projections; complex L2-normalisation on Q, K; cosine score; sigmoid gate with bias initialised to $-\log N$; real-weight $\times$ complex-value aggregation.

3.3.2 complex_screen

With the same complex-linear projections and L2-normalisation as PCT, plus learnable scalar threshold t and per-token magnitude $r_{ij} = \|q_i\| \cdot \|k_j\|$:

$$\begin{aligned} g_{ij} &= r_{ij}^2 \cdot \max(\text{Re}\langle \bar{q}_i, \bar{k}_j \rangle - t, 0)^2 \\ \bar{g}_{ij} &= \text{TanhNorm}(g_{ij}) \\ u_i &= \sum_{j=1}^N \bar{g}_{ij} v_j \\ \text{out}_i &= u_i \odot \text{modReLU}(g_i) \end{aligned}$$

Same overall skeleton as PCT but the smooth sigmoid is replaced by a hard $(s - t)^2$ threshold, plus TanhNorm + modReLU substrate (Nakanishi 2026).

3.3.3 Real-side baselines

- **real_softmax**: standard scaled-dot-product softmax on real Q, K, V (Vaswani 2017).
- **real_sigmoid**: Ramapuram-2024 element-wise sigmoid attention on real $QK^\top$ with bias init $-\log N$.
- **real_screen**: Nakanishi-2026 screening on real $QK^\top$ with TanhNorm post-aggregation and softmax=OFF.

3.3.4 Vanilla complex baseline

- **complex_softmax**: Eilers-Jiang-2023 CAtt – softmax of cosine score magnitude on complex Q, K. The literature default for complex attention.

3.4 Substrate choices

The complex-valued cells use three substrate components alongside the gating activation:

1. Native complex linear for QKV / output projections.
2. cval value path (real $\alpha \times$ complex $V \rightarrow$ complex out).
3. Re-cosine score $s = \text{Re}\langle \bar{q}, \bar{k} \rangle$ for the gate input.

Each is interchangeable on PCT. The complex substrate is **existentially** required, but no specific component carries it irreducibly. This robustness of the substrate strengthens the practical recommendation: practitioners can pick any reasonable parameterisation and obtain equivalent results.

3.5 Tasks

#	Task	Params	Seq length	What it tests
1	Copy Memory (short)	K=10, d $\in$ 200, 500	400–1000	short-range retrieval (sanity)
2	Copy Memory (long)	K=10, d $\in$ 1000, 2000	2000–4000	long-range retrieval, dilution stress
3	NIAH	depth_ratio=0.5, vocab=64, L=2048	2048	purely positional retrieval
4	LRA-ListOps small	depth=2, max_args=3	128	algorithmic reasoning, content-driven
5	LRA-ListOps mid	depth=2, max_args=3	1024	algorithmic + long-range
6	FFT-MNIST	t=16 frames	448	phase-decomposed image classification
7	phase-memory	K=8, delay=30	250	phase-sensitive retrieval
8	multi-pitch	K=16	256	phase-superposition decoding
9	LRA-Pathfinder / Text / Image	32 $\times$ 32 / 4K bytes / CIFAR	1024 / 4096 / 1024	LRA-standard long-range tasks

3.6 Training protocol and fairness

- **Parameter fairness:** $real_{dim} = complex_{dim} \times \sqrt{2} \approx 1.41 \times$. Implementation: real cells use dim=184, heads=4, dim_head=46; complex cells use dim=128, heads=4, dim_head=32. The parameter-matched ablation in Appendix F and the parameter-fair mid-scale sweep both confirm this convention.
- **Batch fairness:** all main-table runs use batch_size=32 effective. An earlier audit showed that batch=8 starves softmax / sigmoid in mid-scale; that point is preserved in Appendix F as the C1-violation control.
- **softmax=OFF for all screening cells.**
- **τ -sweep for screening:** 5 threshold inits $\times$ 1 seed $\rightarrow$ best-of selected for the 5-seed run. Eliminates screening hyperparam disadvantage.
- **Stuck-seed protocol:** $\geq 2/3$ seeds at random $\rightarrow$ expand to N=10 and report median + IQR.
- **LR sweep:** 3 LRs for the H3 robustness claim; main-table uses best LR per cell.
- **Optimisation:** AdamW with weight_decay=1e-5, gradient clipping at 1.0, cosine LR schedule with linear warmup.

3.7 Hardware / reproducibility

H100 80 GB GPUs (primary), with A100 80 GB and consumer-grade GPUs (RTX 3090 / 4090, ≥ 0.99 reliability) for parallel sweeps. Code, configurations, and checkpoints will be released together with the experiment configs needed to reproduce every reported run.

4 Experiments / Results

This section reports the empirical findings of the 6-cell comparison. §4.1 summarises the headline outcome – both predictions of §3.1 confirmed plus two emergent findings. §4.2–§4.8 give the detailed evidence per hypothesis. *Mechanistic interpretation is deferred to §5.*

4.1 Summary: predictions confirmed plus emergent findings

The 6-cell comparison **confirmed both predictions of §3.1** and additionally revealed two findings that extend the picture beyond what was a priori expected:

- **(1) Confirmed – the "complex penalty" of vanilla complex is real.** On non-phase tasks, `real_softmax` outperforms `complex_softmax`:
 - Copy d=500 mid: real **0.45** vs complex 0.10
 - Copy d=2000 mid: real 0.10 vs complex 0.08
 - LRA-ListOps small: real 0.625, complex 0.637 (close)
 - NIAH L=2048 mid: both 0.000

Naive softmax-of-complex does not deliver complex generalisation. Detailed in §4.5.

- **(2) Confirmed – PCT delivers complex computation.** Under parameter-fair comparison ($real_{dim} = complex_{dim} \times \sqrt{2}$), PCT outperforms `real_softmax` on every task tested. The hypothesised "useful complex computation" extracted by PCT is empirically present:
 - Copy d=500 mid: PCT **1.00** vs `real_softmax` 0.45
 - Copy d=2000 mid: PCT **1.00** vs `real_softmax` 0.10
 - NIAH L=2048: PCT **1.00** vs `real_softmax` 0.00
 - LRA-ListOps small: PCT **0.719** vs `real_softmax` 0.625
 - FFT-MNIST t=16: PCT **0.45** vs `real_softmax` 0.32

Detailed in §4.6.

- **(3) Emergent finding – PCT matches or exceeds the strongest real-side baseline on every synthetic and algorithmic task category in the suite.** Beyond outperforming `real_softmax`, PCT also matches or beats `real_screen` on:
 - Copy d=500/d=2000 mid: PCT 1.00 = `real_screen` 1.00 (tie)
 - NIAH L=2048: PCT 1.00 = `real_screen` 1.00
 - LRA-ListOps small: PCT **0.719** > `real_screen` 0.688
 - FFT-MNIST t=16: PCT 0.45 > `real_screen` 0.43
 - phase-memory: PCT 0.99 $\approx$ `real_screen` 1.00
 - multi-pitch: PCT **0.87** > `real_screen` 0.85

The "complex helps only on phase tasks" reading of prior CvNN literature is refuted: PCT matches or exceeds the best real-side cell even on tasks where complex was traditionally considered weak. Detailed in §4.2, §4.3, §4.7.

Calibration counterexample: on the one real-world physical-complex task in our suite – RML2016 wireless I/Q modulation classification at small scale – `real_screen` substantially beats PCT. PCT remains in the top tier but is no longer the leader. We acknowledge this as a counterexample within the phase-sensitive task category and discuss possible mechanisms in §5.9. The headline pattern across all other phase-sensitive tasks is unaffected.

- **(4) Emergent finding – PCT is uniquely robust to hyperparameter variation, beyond**

mere task generality. Beyond varying tasks, varying the LR window and batch regime within long-range Copy reveals a structural robustness that no other cell – real or complex – exhibits:

- LR window 1e-3, 3e-3, 1e-2 on Copy d=1000: PCT 100 % / 100 % / 100 %; all other cells fail at one or more LR points.
- Batch regime b=8, b=32, b=256 on long-range Copy: PCT 1.00 / 1.00 / 1.00; only PCT achieves the full sweep.

PCT is the only cell that requires no LR / batch tuning to deliver. Detailed in §4.4.

The four bullets above frame the §4 narrative. Subsections §4.2–§4.8 give the detailed evidence tables and per-task accuracy values for each hypothesis.

Abbreviation in table headers. To save horizontal space, table headers in §4 use the prefix abbreviations `c_*` and `r_*` for `complex_*` and `real_*` respectively (e.g., `c_softmax` = `complex_softmax`, `r_screen` = `real_screen`). The full cell names are used everywhere else in the text.

4.2 H1 – Headline matrix

Table H1: Cross-task headline matrix. Across the synthetic and algorithmic rows, every row admits at least one of PCT, `complex_screen` at or near the top. The two Real RadioML rows are a calibration counterexample – a phase-sensitive real I/Q domain where `real_screen` substantially beats both PCT and `complex_screen` (discussed in §4.9 / §5.7).

Task	<code>r_softmax</code>	<code>r_sigmoid</code>	<code>r_screen</code>	<code>c_softmax</code>	PCT	<code>c_screen</code>
Copy d=500 (mid)	0.45	0.94	1.00	0.10	1.00	0.45
Copy d=2000 b=32 (mid)	0.10	0.10	1.00	0.08	1.00	0.53
NIAH L=2048 (mid)	0.00	0.00	1.00	0.00	1.00	1.00
LRA-ListOps L=1024 dep=2 (small)	0.146	0.177	0.698	0.104	0.854	0.833
LRA-ListOps small	0.625	0.594	0.688	0.637	0.719	0.715
FFT-MNIST t=16	0.32	0.31	0.43	0.39	0.45	0.45
phase-memory delay=30	0.98	0.96	1.00	0.93	0.99	1.00
multi-pitch	0.81	0.81	0.85	0.82	0.87	0.88
Real RadioML L1	0.30	0.30	0.36	0.24	0.35	0.34
Real RadioML L2	0.22	0.24	0.39	0.27	0.35	0.34

4.3 H2 – NIAH dominance

NIAH is purposefully constructed to deny complex Q, K their phase-coherence inductive bias: needle and distractors are drawn from the same uniform-random vocabulary, so retrieval is purely positional. A priori, this is the regime where vanilla complex attention should be at greatest disadvantage. The relevant empirical question is whether the PCT-family cells *still* compete with real-side baselines. Table H2 below gives the result on NIAH L=2048 mid-scale: both PCT and `complex_screen` solve it deterministically, matching `real_screen` – the only real-side cell that solves it. The four cells without a screening or sigmoid-on-cosine structure all collapse to 0.000.

Table H2: NIAH L=2048 dominance.

4.4 H3 – LR / batch robustness

PCT is the **unique** cell that retains full accuracy across the full LR window (1e-3 / 3e-3 / 1e-2) on small-scale Copy d=1000 *and* across the full batch regime (b=8, 32, 256) on mid-scale long-range Copy. Other cells succeed only inside narrow corridors – `real_sigmoid` reaches 1.00 only at LR=3e-3, `real_screen` only at LR=1e-2 (and is seed-dependent at lower LRs), `complex_screen`

collapses to 0.08 at LR=1e-2 and to 0.03 at b=8 / d=2000.

Table H3 below combines the LR sweep (left half: small-scale, dim=32, batch=256) and the batch sweep (right half: mid-scale, dim=256, depth=6) into a single matrix; the unique all-1.00 pattern of PCT shows up as the all-bold row.

Table H3: LR robustness (Copy d=1000, small-scale) and batch robustness (long-range Copy, mid-scale).

Cell	LR=1e-3	LR=3e-3	LR=1e-2	b=8 d=2000	b=32 d=2000	b=256 d=500
real_softmax	0.37	0.69	0.31	0.04	0.10	0.05
real_sigmoid	0.07	1.00	0.22	0.04	0.10	0.07
real_screen	0.55	0.70	1.00	0.69	1.00	0.79
complex_softmax	0.09	0.39	0.35	0.080	0.08	0.06
PCT	1.00	1.00	1.00	1.00	1.00	1.00
complex_screen	0.39	1.00	0.08	0.03	0.53	0.31

4.5 H4 – vs vanilla complex

PCT wins vs `complex_softmax` (the literature default – Eilers CAtt / Holographic Transformer family) on every task tested. The Real RadioML rows give the **largest single margin in the entire H4 table**: PCT clears `complex_softmax` by +0.10 at L1, establishing that softmax-of-complex is decisively the worst default specifically for physical complex domains where the literature most strongly motivates complex transformers. Table H4 below.

Table H4: PCT vs vanilla complex (`complex_softmax`).

Task	PCT	c_screen	c_softmax
Copy d=500 mid	1.00	0.45	0.10
Copy d=2000 mid	1.00	0.53	0.08
NIAH L=2048	1.00	1.00	0.00
LRA-ListOps small	0.719	0.715	0.637
LRA-ListOps mid (L=1024)	0.854	0.833	0.104
FFT-MNIST t=16	0.45	0.45	0.39
phase-memory	0.99	1.00	0.93
multi-pitch	0.87	0.88	0.82
Real RadioML L1	0.345	0.341	0.244
Real RadioML L2	0.352	0.337	0.273

4.6 H5 – vs vanilla real, parameter-fair

The PCT family beats `real_softmax` on long-range Copy and NIAH by large margins, with the comparison run at parameter-fair dim=184 (so $real_{dim} = complex_{dim} \times \sqrt{2}$). On short-range and phase-sensitive synthetics the gap is smaller, but PCT wins. Table H5 below.

Table H5: PCT vs vanilla real (`real_softmax`), parameter-fair.

Task	PCT	c_screen	r_softmax
Copy d=500 mid	1.00	0.45 (tie)	0.45
Copy d=2000 mid	1.00	0.53	0.10
NIAH L=2048	1.00	1.00	0.00
LRA-ListOps small	0.719	0.715	0.625
FFT-MNIST t=16	0.45	0.45	0.32
phase-memory	0.99	1.00	0.98
multi-pitch	0.87	0.88	0.81
Real RadioML L1	0.35	0.34	0.30
Real RadioML L2	0.35	0.34	0.22

4.7 H6 – vs strongest real-side baseline

real_screen is the strongest real-side cell across most of the suite – the only real cell that solves NIAH L=2048 deterministically and competes on Copy d=2000 mid-scale. Calibrated by the Real RadioML rows of Table H6 below: **on synthetic and algorithmic benchmarks, PCT or complex_screen matches or exceeds real_screen on every task; on Real RadioML L1/L2, real_screen substantially beats both PCT and complex_screen.** We acknowledge Real RadioML as a counterexample within the phase-sensitive task category – see §4.9 for the data and §5.7 for discussion. The PCT-dominance pattern remains intact across all other tasks tested.

Table H6: PCT vs strongest real-side baseline (real_screen).

Task	PCT	c_screen	r_screen
Copy d=500 mid	1.00	0.45	1.00
Copy d=2000 mid	1.00	0.53	1.00
NIAH L=2048	1.00	1.00	1.00
LRA-ListOps small	0.719	0.715	0.688
LRA-ListOps mid (L=1024)	0.854	0.833	0.698
FFT-MNIST t=16	0.45	0.45	0.43
phase-memory	0.99	1.00	1.00
multi-pitch	0.87	0.88	0.85
Real RadioML L1	0.35	0.34	0.36
Real RadioML L2	0.35	0.34	0.39

4.8 Depth scaling

A common concern with complex-valued transformers is **depth scalability** – the worry that complex computation accumulates phase noise across layers, breaking deeper stacks. We probe this concern empirically with PCT at six methodology-aligned depths spanning a $10\times$ param range, plus mid-scale evidence from §4.2 / §4.3 / §4.4.

Depth=4: PCT achieves 1.000 on Copy d=1000, **0.854** on LRA-ListOps L=1024 dep=2, **1.000** on LRA-Text 4K (N=3), **0.458** on LRA-Image, **0.927** on FFT-MNIST t=16, **0.997** on multi-pitch – the headline 7-task matrix of §4.2.

Depth=6:

- Copy d=2000 b=32: **1.000**

- Copy d=2000 b=8: **1.000** (N=3)
- Copy d=5000: **1.000** (N=5)
- NIAH L=2048: **1.000**

6-depth methodology-aligned LRA-ListOps L=1024 sweep

– full per-depth metrics and trajectories in the supplementary
depth-scaling note:

Two clean reads:

1. **best_acc is flat across the 10× depth range.** Linear fit slope = -0.009 , $R^2 = 0.04$. The flat reading is consistent with **lucky-batch saturation at eval_batch=32**, not with depth collapse: the trajectory snapshots show every depth reaches ≥ 0.69 by step 30K and stays there, with no architecture failing to train.

depth	params	best_acc	best_eval_loss	wall
2	0.40 M	0.8125	0.6516	40 min
4	0.80 M	0.7812	0.6562	62 min
6	1.19 M	0.8125	0.6106	86 min
10	1.98 M	0.8125	0.6370	132 min
14	2.77 M	0.8125	0.6154	179 min
20	3.96 M	0.7812	0.5397	250 min

2. **best_eval_loss is weakly monotone with a non-monotonic dip** (d=6 $\rightarrow$ d=10: 0.611 $\rightarrow$ 0.637). 6-point power-law fit: $\text{loss} \approx \text{params}^{-0.065}$, $R^2 = 0.58$.

Caveats:

- eval_batch=32 produces 0.0625 quanta and an apparent ceiling at $13/16 = 0.8125$; an n=2048 stable evaluation on the saved checkpoints is the cleanest fix and is queued.
- An independent archive evaluation at batch=16 with n=2048 stable evaluation on consumer hardware found accuracy plateauing at 0.50–0.56 across d=4, 6, 10 with no significant depth signal – consistent with the 6-point flat best_acc here once lucky-batch saturation is removed.

The combined empirical picture:

- On 7 synthetic / algorithmic / classification tasks tested at depth=4 and at depth=6, PCT reaches 1.000 (perfect) on long-range Copy and on NIAH L=2048 – *none of these collapse with depth*.
- On the dedicated 6-point LRA-ListOps L=1024 sweep, PCT trains without instability or accuracy collapse at every depth from 2 to 20. The signal on whether deeper depths *help* accuracy is buried under measurement noise and remains open; the signal that deeper depths *do not hurt* is clean.
- The depth=20 architecture trains cleanly with the standard recipe.

Summary: PCT operates without depth-related accuracy collapse across all depths we tested. We do **not** fit a quantitative scaling law: the 6-point best_acc is flat at the lucky-batch saturation ceiling, and the 6-point best_eval_loss fit ($\alpha = -0.065$, $R^2 = 0.58$) is too weak to support a Chinchilla-style scaling claim. The empirical claim is qualitative and conservative: **PCT does not exhibit a depth-collapse failure mode at any depth we tested.**

4.9 Real-domain validation

To check that the synthetic and algorithmic dominance pattern carries over to physical complex-domain data, we benchmark all six cells on (a) the public **RML2016** mirror of RadioML and (b) a 10-piece test split of **MusicNet**. Both are small-scale runs at a dim=64, depth=3 configuration with N=3 seeds per cell. Full setup and per-seed numbers in the supplementary real-domain note.

Real RadioML:

Two clean signals from the table:

- **real_screen** substantially beats **PCT** and **complex_screen** on both L1 and L2. On L2 the margin exceeds the seed std for all three cells. **This is a phase-sensitive task on which PCT loses to a real cell** – the only such task in our benchmark suite. We discuss this explicitly in §5.9 as a counterexample within the phase-task category that the closeness-to-PCT framing does not yet explain.
- **complex_softmax** is the worst cell on the I/Q domain at L1. PCT (0.345) clears **complex_softmax** by +0.10. This is the cleanest evidence in the paper that **softmax-of-complex inheritance from real-side practice is a poor default specifically for physical complex domains**, where complex transformers are most strongly motivated.

Cell	L1	L2	params
real_softmax	0.296	0.224	149.7 K
real_sigmoid	0.303	0.241	149.7 K
real_screen	0.363	0.389	163.0 K
complex_softmax	0.244	0.273	150.4 K
PCT	0.345	0.352	150.4 K
complex_screen	0.341	0.337	163.6 K

Real MusicNet:

Real MusicNet at this configuration is **data-ceiling-bound and is reported for transparency, not as a comparative result**. L1 cell separation is 0.02 F1 – within seed variance. L2 saturates: all six cells converge to F1 = 0.143. Followups are out of scope for this paper.

Cell	L1	L2
real_softmax	0.201	0.143
real_sigmoid	0.201	0.143
real_screen	0.214	0.143
complex_softmax	0.201	0.143
PCT	0.203	0.143
complex_screen	0.218	0.143

4.10 Notes on the choice of effective batch size, and data availability

On batch_size = 32 effective as the main comparison setting. We use `batch_size = 32` effective as the headline batch size throughout §4 because (i) it is the most commonly used effective batch size in the recent attention literature at this model scale, and (ii) it stays within the memory envelope where no out-of-memory failure occurred for any of the six cells in our experiments. We also ran extensive comparisons at `batch_size = 8`, but `b=8` is not a practically informative setting outside of strictly memory-constrained deployments, so we do not adopt it as the headline comparison. We note for the record that **at batch_size = 8 the PCT family showed an even more pronounced cross-task tendency to be the unique top-ranked cell** than at `batch_size = 32` – i.e., PCT’s batch robustness, already noted in §4.4 (H3), strengthens further in this regime.

Data availability. All raw results underlying the tables and figures in §4 are available at:

<https://github.com/leohio/phase-coherent-transformer-r-d/tree/main/result>

5 Discussion

This section interprets the §4 results, gives the mathematical framing of why PCT wins, brings in additional cells for mathematical comparison, presents the anti-phase deletion isolation experiment as direct evidence, and discusses scalability to deeper stacks.

5.1 Why PCT wins: token non-competition for multi-layer phase preservation

The PCT family’s empirical advantage stems from two structural properties of the gate:

(a) **Token non-competition** – the gate is element-independent, with no row-normalisation. Phase superposition of complex value contributions is preserved without competition between tokens. `complex_softmax` violates this property and exhibits long-range dilution failure.

(b) **Multi-layer phase preservation** – the gate is smooth on the full real domain, in particular gradient-positive on the negative-cosine-score region, so anti-phase contributions to the value aggregate are not deleted at any layer. `complex_relu` violates this property fully and `complex_screen` partially.

These two properties define **two-level phase coherence**: per-layer (L1) preservation through token non-competition (a), and all-layer cascade (L2) preservation through multi-layer phase preservation (b). Cells deviate from PCT in three structurally distinct grades:

- **Partial deviation:** `complex_screen` violates (b) only below a learned threshold; smooth above. Result: L2 partial $\rightarrow$ task-conditional behaviour.
- **Bypassed by the substrate:** `complex_softplus` is C2-violating on $\mathbb{R}$ but the L2-normalised cosine score restricts the operating range to $[-\text{sqrt}d, \text{sqrt}d]$ where softplus is bounded ($M \approx 4$). Result: practically equivalent to PCT, 1.000 on Copy d=1000.
- **Strict in operating range, not bypassed:** `complex_relu` and `complex_clamped_relu` $\rightarrow$ L2 cascade fully fails $\rightarrow$ chance level on long-range retrieval. `complex_cubic` $\rightarrow$ cascade severely degraded $\rightarrow$ 0.200 on Copy d=1000.
- **Token-competition (C4 $\times$):** `complex_softmax` fails (a) $\rightarrow$ fails L1 $\rightarrow$ catastrophic on long-range retrieval. This is a separate failure mechanism from the C3-cascade collapse.

The empirical separation on Copy d=1000 (N=3) is clean: cells whose deviations are *partial or bypassed* match PCT; cells whose deviations are *strict in the operating range* fail. See §5.4 for the full 2×2 isolation matrix.

5.2 Two-level phase-coherence and the 4-condition framework

Formally, the two-level phase-coherence property and the four-condition framework on the gate $\alpha = f$ are:

Two-level phase coherence:

- **(L1) Per-layer phase coherence** – Definition 1: an attention layer preserves the phase relationships of inputs at each layer.
- **(L2) All-layer cascade phase stability** – Definition 3: a stack of L per-layer phase-coherent layers preserves a phase invariant L-independently – accumulated phase noise stays $O(1)$ across depth.

Four conditions on the gate:

- **C1** – real-valued gate $\alpha \in \mathbb{R}$,
- **C2** – bounded gate, in the **operating-range** sense: $|f| \leq M$ for $s \in [-\sqrt{d}, \sqrt{d}]$.
- **C3** – smooth, gradient-nonzero on the **operating range**,
- **C4** – element-independent gate.

Operating-range form vs strict-on- $\mathbb{R}$ form of C2 / C3 is a *load-bearing distinction*. The architectural substrate **softens strict-on- $\mathbb{R}$ violations into operating-range violations**:

- A cell that is C2-violating on $\mathbb{R}$ but bounded on the operating range **satisfies the operating-range form of C2** and is empirically close to PCT.
- A cell whose violation persists in the operating range **fails C2 in the operating-range sense** and exhibits cascade degradation.

Whenever we say "C_k satisfied" below, we mean it in the operating-range form unless explicitly noted.

Two key results:

- **Theorem 1: C1 & C4 $\Rightarrow$ L1.** Elementary proof via complex-linear conjugate cancellation. PCT and `complex_screen` satisfy L1; `complex_softmax` does not.
- **Theorem 2: C1 & C2 & C3 & C4 & substrate non-expansion $\Rightarrow$ L2,** with C2 / C3 read in the operating-range sense. Proof via global-mode decomposition & linearised Jacobian & Doeblin contraction on the zero-mean phase subspace & substrate non-expansion. The proof’s M constant is the operating-range bound, and the empirical results in §5.4 confirm the necessity of all four conditions in their operating-range form. Residual pieces detailed in the companion document.

This framework grounds the closeness-to-PCT axis as a structural taxonomy of *how* a cell deviates:

- **PCT:** all four conditions strict $\checkmark$ in operating range, also strict on $\mathbb{R}$ – tightest L2 constants. Theorem 2 applies cleanly.
- **Close to PCT with bypassed deviation:** C2 violated on $\mathbb{R}$ but operating-range form satisfied ($M \approx 4$). Theorem 2 still applies with a slightly looser constant. Empirically equivalent to PCT.
- **Close to PCT with partial deviation in the gate domain:** C3 satisfied above the threshold and violated below. Theorem 2 holds *for inputs that stay above the threshold*; partial cascade collapse for inputs crossing the threshold.
- **Far from PCT with strict C3 violation:** C3 $\times$ on a non-trivial subset of the operating range. Theorem 2’s Lemma B linearisation breaks, the cascade summation does not close, and accuracy collapses to chance.
- **Far from PCT with strict C2 violation in operating range:** C2 $\times$ with M much larger than the substrate’s smoothing constant. Theorem 2’s M constant grows polynomially in cascade depth, geometric series fails to contract, partial cascade collapse.
- **Non-PCT:** C4 violation. L1 fails before any cascade analysis is needed – Theorem 2 is moot, and the failure mode is row-normalisation dilution rather than cascade noise.

This subsection presents the framework without proofs; full statements and proofs are in Appendix M.

5.3 Designed counterexamples through the C1–C4 lens

Five additional cells, brought in here for mathematical discussion, span the C2 $\times$ C3 design space and illuminate the framework:

Note. The cells discussed in this subsection – including those that violate one or more PCT conditions – are *not* broken architectures. ReLU, softplus, tanh, and clamping are all standard, well-established gating primitives that yield perfectly functional Transformers on the real-valued side. The point of this analysis is not that these gates are bad in general; it is that, **once lifted to a complex-valued substrate**, the structural conditions C1–C4 become load-bearing for cascade phase coherence, and gates that fail one of these conditions exhibit the specific failure modes predicted by the framework. The same primitives, used in a real Transformer, would not show these pathologies.

- **tanh+1:** identical to sigmoid up to gain scaling ($\tanh(s) + 1 = 2\sigma(2s)$). Predicted to behave equivalently to PCT on phase-memory and FFT-MNIST, where it does match sigmoid. On long-range Copy, however, `complex_tanh1` exhibits stuck-seed pathology that sigmoid does not – within the PCT family the doubled gradient gain makes optimisation more brittle on cascade-stress tasks, so we treat sigmoid as the canonical PCT instance.
- **softplus:** gate gradient is sigmoid, which is bounded and nonzero everywhere – so anti-phase preservation (C3) is intact. The gate output is unbounded above on $\mathbb{R}$, but the L2-normalised cosine score restricts the operating range to $[-\text{sqrt}d, \text{sqrt}d]$, on which softplus is bounded ($M \approx 4$ with $d=128$, $N=1021$). Predicted to behave close to PCT despite the strict-on- $\mathbb{R}$ unboundedness. Confirmed: 1.000 $\pm$ 0.000 on Copy $d=1000$ (§5.4).
- **cubic:** Taylor expansion of `sinh` truncated at the cubic term. Strict C3. C2 strongly violated: in

operating range $[-\sqrt{d}, \sqrt{d}]$, the gate output reaches $\sqrt{d} + d^{1.5}/6 \approx 252$ for $d=128$. Predicted: severe degradation due to magnitude divergence in cascade, but not full collapse since C3 is intact. Confirmed: 0.200 ± 0.030 on Copy $d=1000$ (§5.4).

- **clamped_relu**: hard sigmoid above zero. C2 strict. C3 fully violated. Predicted: full collapse to chance – anti-phase deletion is fatal regardless of M. Confirmed: 0.103 ± 0.039 on Copy $d=1000$ (§5.4).
- **ReLU**: gate gradient is zero on the negative half-domain – full anti-phase deletion. Predicted catastrophic on phase-sensitive and long-range tasks. Confirmed: phase-memory acc 0.121 (N=3 random); §5.4 isolation gives Copy $d=1000$ acc 0.107 (N=3 random) – same chance level as clamped_relu, confirming C3 dominates over C2 once C3 is violated.

The five cells together fill the 2×2 design space $C2 \checkmark/\times \times C3 \checkmark/\times$; their empirical behaviour precisely matches the C1–C4 framework’s predictions, with C3 violation dominating over C2 violation when both are violated. Together with PCT (sigmoid), **complex_screen**, and **complex_softmax**, these cells span the closeness-to-PCT axis, and their empirical behaviour matches the C1–C4 predictions.

Cell	C1	C2	C3	C4
tanh+1	✓	✓	✓	✓
softplus	✓	partial	✓	✓
cubic	✓	×	✓	✓
clamped_relu	✓	✓	×	✓
ReLU	✓	partial	×	✓

5.4 Two-axis ($C2 \times C3$) isolation experiment

Task: copymem $K=10$ $d=1000$. **Architecture:** $\text{dim}=128$, $\text{depth}=4$, $\text{heads}=4$, $\text{dim_head}=32$, $\text{ff_mult}=4$, $\text{batch}=32$, $\text{lr}=3e-3$, $\text{steps}=2000$, AdamW. **N=3** seeds each.

We extend the 2-cell isolation to a **complete 2×2 design** by introducing two new cells that selectively violate one of C2, C3 while keeping the other strictly satisfied:

- **complex_cubic** with gate $f(s+b) = (s+b) + (s+b)^3/6$. **C3 strictly satisfied** ($f' = 1 + (s+b)^2/2 \geq 1$ everywhere), **C2 strongly violated** in operating range ($M \approx 252$ for $d=128$).
- **complex_clamped_relu** with gate $f(s+b) = \text{clamp}$. **C2 strictly satisfied**, **C3 fully violated**.

Headline 2×2 matrix:

	C2 ✓	C2 ×
C3 ✓	complex_sigmoid: 1.000 complex_softplus: 1.000	complex_cubic: 0.200
C3 ×	complex_clamped_relu: 0.103	complex_relu: 0.107

Three independent claims fall out:

1. **C3 is empirically necessary for the cascade** (strict). **complex_clamped_relu** collapses to chance level (0.103 ± 0.039), eval loss flat at $\log V$. Anti-phase deletion alone – with the gate output bounded identically to sigmoid – destroys the ability to learn long-range retrieval. The N=3 per-seed values match **complex_relu**’s within seed noise. **Once C3 is violated, C2 status is irrelevant.**
2. **C2 is also necessary in its operating-range form, when M is large enough.** **complex_cubic** ($C2 \times M \approx 252$, $C3 \checkmark$ strict) drops from 1.000 to **0.200 ± 0.030** – an 80 % accuracy loss. Even with the gate gradient strictly positive everywhere ($f' \geq 1$), the unbounded gate output causes the cascade to fail. **C2 violation alone is a real failure mode.**
3. **Magnitude of C2 violation matters monotonically.** **softplus** ($M \approx 4$) achieves 1.000; **cubic** ($M \approx 252$) achieves 0.200. Somewhere between $M=4$ and $M=252$ lies a transition. **C2 is not a binary on/off condition – it’s a magnitude-dependent factor** that interacts with cascade depth.

An earlier interpretation, based on softplus alone ($M \approx 4$ in the operating range), had argued that "C2 is automatic given L2-normalisation + a continuous gate" and could be dropped from the explicit

four-condition framework. **That simplification is withdrawn:** at sufficient M, C2 violation is fatal.

The four-condition framework is therefore correctly stated as $C1 \& C2 \& C3 \& C4 \Rightarrow L2$, with all four conditions independently necessary in their operating-range form.

The clean reading of the 2×2 matrix:

- **Top-left:** the cell PCT was designed to be – succeeds.
- **Top-right:** partial collapse, still some learning.
- **Bottom-left:** full collapse to chance.
- **Bottom-right:** full collapse – but no worse than C3-only, confirming C3 dominates the failure.

Why C3 violation dominates over C2: at the linearised per-layer Jacobian, C3 violation introduces a discontinuity in the gate gradient that breaks the linearisation entirely; the cascade-summation argument of Theorem 2 cannot recover. C2 violation degrades the geometric series’s contraction constant but does not break linearisation – so C2 violation produces partial degradation, while C3 violation produces structural failure. The empirical 0.200 vs 0.103 gap matches this prediction.

(Mathematical formalisation of why anti-phase deletion drives cascade collapse – i.e., why it breaks all-layer phase coherence (L2) – is in the companion document: the linearised per-layer Jacobian requires gate differentiability, which both ReLU’s hard cutoff and clamped_relu’s two-sided saturation destroy, breaking the geometric-summation argument of Theorem 2.)

5.4.1 Confirmation across additional tasks

The headline 2×2 matrix above uses Copy d=1000 as the cleanest single-task probe. The same isolation experiment also covered five additional tasks, and the collapse pattern carries across them. We report `complex_cubic` and `complex_clamped_relu` only, since `complex_sigmoid` is the matched-config control covered by the headline matrix.

Task	c_clamped_relu	c_cubic (C3 ✓ strict, C2 × M≈252)	PCT (sigmoid) reference	Pattern
copymem K=10 d=100	0.103	0.200	1.000	distance-independent collapse
copymem K=10 d=200	0.103	0.200	1.000	distance-independent collapse
copymem K=10 d=500	0.103	0.200	1.000	distance-independent collapse
copymem K=10 d=1000	0.103	0.200	1.000	(the headline 2×2 row)
FFT-MNIST t=16	0.323	0.292	0.45	partial degradation
multi-pitch K=16	0.812	0.861	0.997	clamped_relu trivial-baseline collapse

(All values: N=3 seeds ($s \in 0, 1, 2$), identical isolation architecture and training as the headline 2×2 matrix above. PCT reference values are taken from §4.2 / §4.8 – they are not re-runs at the isolation lr/steps configuration but the qualitative ranking is unaffected.)

Three structural observations from the wider task set:

1. **The C3-violation collapse is distance-independent on Copy.** `complex_clamped_relu` reaches exactly the same 0.103 ± 0.039 at every distance $d \in 100, 200, 500, 1000$ – bit-identical to within

numerical noise. The cell does not "almost solve $d=100$ and fail at $d=1000$ "; it never starts learning. Anti-phase deletion alone makes the retrieval task structurally invisible to the model regardless of the inter-token distance, which is exactly what the cascade argument of Theorem 2 predicts: a full C3 violation breaks the per-layer linearisation independently of how many positions the cascade must traverse.

2. **The C2-violation partial collapse is also distance-independent on Copy.** `complex_cubic` returns 0.200 ± 0.030 at every $d \in 100, 200, 500, 1000$, again bit-identical. The cell extracts a small fixed amount of structure regardless of distance, consistent with a cell whose cascade contraction constant is degraded by a fixed multiplicative factor rather than by a distance-dependent decay.
3. **The C3-violation collapse takes a different form on structured-multilabel tasks.** On multi-pitch $K=16$, `complex_clamped_relu` returns 0.812 ± 0.000 on all three seeds. 0.812 is the trivial baseline $/K = 13/16$ recovered by predicting "no pitch active" everywhere; $\text{std}=0$ across independent random initialisations is a strong signature that the cell never engages with the task structure. `complex_cubic`, in contrast, sits at 0.861 – slightly above trivial, consistent with the partial-degradation pattern seen on Copy.

The wider table thus reinforces the headline finding rather than complicating it: **C3 strict violation produces structurally identical "no-learning" failure across retrieval-style and structured-multilabel tasks, while C2 strict violation produces a smaller, distance- and difficulty-independent partial drop.** The cleanest single-task probe is Copy $d=1000$, which is why the headline 2×2 matrix uses it as the canonical entry point.

5.5 Scalability: phase coherence resolves the depth concern for complex transformers

A widely-held concern in the complex-NN literature is depth scalability: complex transformers are feared to fail at greater depth because phase information accumulates noise across layers, breaking the cascade. The two-level phase-coherence framework gives a structural answer to this concern, and our depth-scaling experiments (§4.8) provide empirical confirmation.

The structural answer: PCT’s all-layer cascade phase stability ensures that per-token phase perturbations propagate through depth L with an L -independent Lipschitz constant. The cascade does *not* compound phase noise; the Doeblin contraction on the zero-mean phase subspace closes the geometric series. Cells lacking L2 instead accumulate phase noise as $O(L)$, predicting and matching the empirical collapse on long-range tasks.

The empirical confirmation spans depths 2, 4, 6, 10, 14, 20 – a $10 \times$ range – on a methodology-aligned LRA-ListOps $L=1024$ sweep, plus mid-scale evidence at depths 4 and 6 on the seven §4.2 tasks:

- **Depth=4:** PCT solves Copy $d=200/500/1000$ at 1.000 across the full LR window; LRA-ListOps $L=1024$ $\text{dep}=2$ at **0.854** ($N=3$); LRA-Text 4K at 1.000 ($N=3$); FFT-MNIST at 0.927; multi-pitch at 0.997.
- **Depth=6:** PCT achieves 1.000 ($N=3$) on Copy $d=2000$ $b=32$, 1.000 ($N=5$) on Copy $d=5000$, and 1.000 on NIAH $L=2048$. No depth-related performance collapse.
- **6-point LRA-ListOps $L=1024$ sweep:** PCT trains cleanly at every point in 2, 4, 6, 10, 14, 20, with `best_acc` clustered at the lucky-batch saturation ceiling (4 of 6 points at $13/16 = 0.8125$, slope = -0.009 vs $\log(\text{params})$, $R^2 = 0.04$). `best_eval_loss` is weakly monotone ($\alpha = -0.065$, $R^2 = 0.58$, 6 points) with a non-monotonic dip at $d=10$. $\text{depth}=20$ jobs train without instability – a strong signal against the depth-collapse worry, even though the data is too noisy to support a quantitative scaling law.

We are careful not to overclaim. The eval-batch saturation hides the true plateau, and the 4-point fit’s earlier $R^2 = 0.71$ dropped to 0.58 once $d=6$ and $d=10$ were added in the same $\text{batch}=32$ config. The empirical evidence supports the **conservative** claim that depth does not break PCT, **not** the strong claim that depth produces a clean Chinchilla-style scaling law. The cleanest follow-up is an $n=2048$ stable evaluation on the saved checkpoints.

The structural framework predicts this conservative claim cleanly: Theorem 2’s L-independent cascade Lipschitz constant rules out architecture-induced depth collapse, but says nothing about whether deeper architectures *help* on a fixed task – that is a task-and-data question outside the framework’s scope.

The empirical claim therefore reads: **PCT scales without depth-related accuracy collapse across all depths we tested**, and the framework gives a structural reason to expect this generalises. The standard worry that complex transformers do not scale to deeper architectures is, within our benchmark scope, dissolved by phase-coherent attention.

5.6 PCT vs complex_screen: when does each suffice?

In the closeness-to-PCT taxonomy of §1, **complex_screen** is *close to PCT* by virtue of a **partial** C3 deviation, and **complex_softplus** is *close to PCT* by virtue of a **bypassed** C2 deviation. Both retain L2 in a qualified form: softplus retains L2 fully, while screen retains L2 only on inputs that stay above the threshold.

Read through the two-level coherence framework: tasks that probe stack-level coherence under heavy loose-tuning conditions – long-range Copy under loose LR/batch, where every other cell loses corridor robustness (§4.4) – expose **complex_screen**’s **partial** L2 deficit, and PCT wins. Tasks where per-layer coherence (L1) plus a clean threshold is enough admit **complex_screen** and **real_screen** as deterministic solvers tying with PCT; PCT also solves NIAH L=2048 deterministically (1.000) under our mid-scale config, so the smooth gate is not penalised on this regime. The two architectures are therefore complementary specialists within the phase-coherent family – **complex_screen** matches PCT on L1-sufficient tasks where its partial C3 deviation does not engage, while PCT additionally retains accuracy in regimes that probe the cascade.

This generalises: any cell whose deviation is *partial* or *bypassed* sits in the close-to-PCT regime and produces strong benchmarks. The closeness-to-PCT axis is therefore not just a heuristic ordering but a structurally grounded taxonomy with predictive empirical content.

5.7 Real RadioML – a phase-task counterexample we do not yet explain

The matrix of §4.2 and the task-by-task comparison in §4.7 single out **Real RadioML L1/L2** as the one task in our suite where **real_screen** substantially beats both PCT and **complex_screen**: **real_screen** 0.363 / 0.389 vs PCT 0.345 / 0.352 vs **complex_screen** 0.341 / 0.337. The L2 margin clearly exceeds the seed std for all three cells. This is the only task in our suite that exhibits this pattern.

Crucially, **Real RadioML is a phase-sensitive task**, so the closeness-to-PCT framing would have predicted PCT to be at least competitive with **real_screen**, not behind. Yet on other phase-sensitive tasks in the suite – phase-memory delay=30, multi-pitch – PCT and **complex_screen** are at or near the top, so this is not a category-level failure of the framework. We flag the mechanism behind the Real RadioML gap as an open question and as the most important pending validation.

What this *does not* change: PCT’s dominance on the 7 synthetic + algorithmic tasks of §4.2 is unaffected, and the cleanest signal from Real RadioML is *positive* for the central anti-softmax claim – **complex_softmax** is the worst cell at L1, giving the clearest evidence in the paper that softmax-of-complex inheritance is a poor default specifically for physical complex domains.

6 Conclusion

Our experiments provide strong evidence that complex-valued neural networks can exhibit generalisation beyond explicitly complex-valued signal domains when their attention mechanism is designed to preserve phase information across layers. We propose the **Phase-Coherent Transformer (PCT)** – a complex-valued attention layer with a real-valued, element-independent, smooth gate on the L2-normalised cosine score – as a strong general-

purpose architecture for complex-valued transformers. We propose the **Phase-Coherent Transformer (PCT)** – a complex-valued attention layer with a real-valued, element-independent, smooth gate on the L2-normalised cosine score – as a strong general-purpose architecture for complex-valued transformers. We identify its two core structural properties – **token non-competition** and **multi-layer phase preservation** – as central to generalisation in complex neural networks, and operationalise them as a four-condition framework C1–C4 on the gate. Across the long-range memory, positional retrieval, algorithmic, classification, and phase-sensitive tasks tested under parameter-fair conditions, PCT matches or exceeds every baseline tested, including the strongest real-side non-softmax cell. **Structural proximity to PCT correlates monotonically with generalisation capacity**, and we ground this proximity in the *kind* of deviation a cell has from C1–C4: cells whose deviations are **partial within the gate domain** or **bypassed by the substrate** remain close to PCT and follow it on most benchmarks; cells whose deviations are **strict within the operating range** sit far from PCT and exhibit chance-level or partial collapse. The non-PCT `complex_softmax` collapses on long-range retrieval through a separate mechanism. A complete N=3 two-axis (C2 $\times$ C3) isolation singles out the failure modes: anti-phase deletion (C3) is the dominant axis, strict in-operating-range C2 violation is a secondary failure mode, and all four conditions are independently necessary in their operating-range form. Depth-scaling experiments show no depth-related collapse across the depths tested. Within our benchmark scope, the empirical and structural evidence converges on a single conclusion: **PCT articulates the correct default direction for complex-valued transformers**, and the closeness axis is a constructive guide to which deviations a practitioner can tolerate.

Figures

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2	Activation gradient comparison (PCT σ , <code>complex_screen</code> (s-t) ² +TanhNorm, softmax, ReLU, softplus) on the full domain	§5.3
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4	NIAH L=2048 needle_acc bar chart, <code>complex_screen</code> and <code>real_screen</code> at 1.0, others at 0.0	§4.3
5	LR robustness chart: Copy d=1000 accuracy across LR $\in$ 1e-3, 3e-3, 1e-2, PCT only fully shaded	§4.4
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7	Direct vanilla-complex comparison: PCT family vs <code>complex_softmax</code> across tasks	§4.5
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11	PCT vs <code>complex_screen</code> specialisation diagram: content/phase axis $\times$ positional/threshold axis	§5.6
12	Two-axis (C2 $\times$ C3) isolation 2 $\times$ 2 matrix: PCT 1.000 / cubic 0.200 / clamped_relu 0.103 / ReLU 0.107	§5.4

Appendix M: Mathematical framework – intuitive overview

This appendix gives the definitions, theorem statements, and proof strategies for the two-level phase-coherence framework. Full proofs with explicit constants and a Lean machine-checked formalisation of Theorem 1 are in the companion document at <https://github.com/leohio/phase-coherent-transformer-r-d/tree/main/lean>.

Notation aliasing with the companion document. This appendix uses compressed numbering: Theorem 1 (per-layer phase coherence) and Theorem 2 (all-layer phase coherence) below correspond, in

the companion document and the Lean formalisation, to **Theorem 1** and **Theorem 5** respectively. The intervening indices (Theorem 1', Conjecture 5, and Definitions 2–4) are taken by intermediate definitions and lemmas elaborated in full only in the companion. References to **Lemma A, B, C, D** below are consistent across this appendix and the companion.

M.0 Setting

A token sequence is $X = (x_1, \dots, x_N) \in \mathbb{C}^{N \times d}$. Two phase-perturbation operators play a role:

- **Global phase shift** $R(\varphi) : X \mapsto (e^{i\varphi}x_1, \dots, e^{i\varphi}x_N)$ – same phase applied to every token.
- **Per-token phase shift** $P(\varepsilon) : X \mapsto (e^{i\varepsilon_i}x_i)_i$ – independent phases per token.

A complex attention layer A_θ is parameterised by complex-linear maps W_q, W_k, W_v, W_o and a real-valued gate $f : \mathbb{R} \rightarrow \mathbb{R}$ acting on the cosine score $s_{ij} = \text{Re}\langle \bar{q}_i, \bar{k}_j \rangle$ of L2-normalised queries and keys.

M.1 Per-layer phase-coherence (L1)

Definition 1 (per-layer phase-coherent). A layer A is *per-layer phase-coherent* if (a) it commutes with the global phase shift $R(\varphi)$, and (b) each gate value α_{ij} depends only on the pair (x_i, x_j) – no cross-token coupling through other tokens.

Condition (a) says global phase is a passive labelling, not destroyed by the layer. Condition (b) is the formal version of "no row-norm coupling": softmax violates it, sigmoid does not.

Theorem 1 (C1 & C4 $\Rightarrow$ L1). A layer with a real-valued (C1) elementwise gate (C4) on the cosine score $\text{Re}\langle \bar{q}_i, \bar{k}_j \rangle$ satisfies Definition 1.

Idea. The cosine score is invariant under joint global rotation, so the gate output is invariant; the value path $W_o \sum_j \alpha_{ij} v_j$ is complex-linear, so the global phase factor passes through. (Full elementary proof in the companion appendix.)

Corollary. PCT (sigmoid gate) and `complex_screen` are L1-coherent. `complex_softmax` is **not** L1-coherent: the softmax denominator couples tokens, violating (b).

M.2 All-layer phase-coherence (L2)

L2 captures *cascade* phase stability: stacking L layers should not amplify per-token phase noise.

Definition 2 (cascade phase stable). A composition $A_L \circ \dots \circ A_1$ is *cascade phase stable* if there exist constants $C_0, C_1 > 0$, **independent of L** , such that for every input X and per-token phase perturbation ε with $\|\varepsilon\|_\infty \leq \delta$,

$$\|(A_L \circ \dots \circ A_1)(P(\varepsilon)X) - (A_L \circ \dots \circ A_1)(X)\|_2 \leq C_0\delta + C_1\delta^2.$$

The L -independence is the substantive requirement: an exponential bound $(1 + \gamma)^L \delta$ would not qualify – phase noise would compound with depth.

Definition 3 (all-layer phase-coherent). A stack is *all-layer phase-coherent* if every layer is L1-coherent and the composition is cascade phase stable.

M.3 Theorem 2: L2 sufficiency – proof strategy

Theorem 2 (sketch). Under (C1) & (C3) & (C4), L2-normalisation on Q, K, a continuous gate on the bounded operating range, and a non-expansive substrate (residual & RMSNorm & FFN), the stack is cascade phase stable with L -independent constants.

The proof decomposes the per-token phase perturbation $\varepsilon \in \mathbb{R}^N$ into two orthogonal modes and bounds each:

1. **Global-mode pass-through (Lemma A).** Write $\varepsilon = \bar{\varphi} \cdot \mathbf{1} + \delta$ with $\bar{\varphi}$ the average and $\delta \perp \mathbf{1}$ the zero-mean residual. By L1, the global mode passes exactly through every layer; the cascade error from $\bar{\varphi}$ is $O(\bar{\varphi}) \cdot \|\text{output}\|$, L -independent.
2. **Per-layer linearisation on zero-mean (Lemma B).** A single layer’s effect on δ is, to first order, a Jacobian J_l on $\mathbb{R}^N$ determined by the row-stochasticised gate $P_{ij} = \alpha_{ij} / \sum_k \alpha_{ik}$ and $\eta_{ij} = -\text{Im}\langle \bar{q}_i, \bar{k}_j \rangle$.
3. **Doeblin contraction (Lemma C).** Under attention diffuseness – every gate value bounded below by $\mu \cdot \pi_j$ for a positive distribution π – the row-stochastic matrix P_l satisfies a Doeblin condition with constant $\mu_D > 0$. The operator norm of J_l on the zero-mean subspace is $\leq 1 - \mu_D < 1$.
4. **Geometric cascade summation.** The depth- L Lipschitz on the zero-mean subspace is bounded by $\sum_{l=0}^{L-1} (1 - \mu_D)^l \leq 1/\mu_D$, L -independent.
5. **Substrate non-expansion (Lemma D).** Residual + RMSNorm + FFN is non-expansive on bounded-norm inputs by standard transformer-stability bounds.

M.4 Why anti-phase deletion (C3 violation) breaks the cascade

The linearisation step (2) **requires the gate to be differentiable** – exactly the C3 condition. ReLU has a hard cutoff at zero; clamped-ReLU saturates at both tails. Both delete the gradient signal on a non-trivial subset of the cosine-score domain. Lemma B’s first-order expansion does not hold uniformly through the discontinuity; the geometric series of step 4 does not close. The framework therefore predicts cascade failure for these cells, which matches the empirical chance-level collapse on long-range Copy and the trivial-baseline collapse on multi-pitch in §5.4.

M.5 What is rigorously proven and what is open

- **Theorem 1 (C1 & C4 $\Rightarrow$ L1)** is fully proven and machine-checked in Lean.
- **Lemmas A and B** are fully proven (Lemma B’s second-order remainder is bounded but not tightened).
- **Lemma C** uses the standard Doeblin coupling argument (Levin, Peres & Wilmer 2017, Theorem 4.9).
- **Theorem 2** is drafted modulo two engineering verifications:
 1. The attention-diffuseness condition is preserved across training (at initialisation it holds with $\mu \geq e^{-1}/N$ from the $b = -\log N$ bias; preservation across gradient updates is a conjecture supported by inspection of trained checkpoints).
 2. A bounded-input fixed-point argument that the cascade stays in the regime where the Doeblin contraction dominates the residual diagonal.

Neither residual piece is a deep open problem; both are tractable engineering verifications. The companion appendix contains the full quantitative statements.

Appendix 0. List of tests conducted

The following tests support the headline matrix and the per-hypothesis evidence in §4 / §5. Each cell $\times$ task combination listed below has at least one trained checkpoint and a per-seed metrics record in the public data archive.

Headline 6-cell comparison:

- Copy Memory K=10 at $d \in 100, 200, 500, 1000, 2000, 5000$
- NIAH L=2048
- NIAH L=1024
- LRA-ListOps small-scale (L=128) – synthetic generator with `max_depth=2`, `max_args=3`
- LRA-ListOps mid-scale (L=1024) – same generator, `dim=128` / `depth=4`
- LRA-ListOps standard – for depth scaling
- LRA-Text 4K
- LRA-Image
- FFT-MNIST at $t \in 8, 16$
- phase-memory K=8 delay=30
- multi-pitch K=16 (synthetic) and K=8
- Real RadioML L1 and L2
- Real MusicNet L1 and L2

Two-axis ($C2 \times C3$) isolation experiment – cells `complex_sigmoid`, `complex_softplus`, `complex_cubic`, `complex_clamped_relu`, `complex_relu`; task `copymem` K=10 $d=1000$; N=3 seeds ($s \in 0, 1, 2$). Detailed in §5.4.

LR / batch robustness sweeps:

- LR sweep on Copy $d=1000$ at $LR \in 1e-3, 3e-3, 1e-2$
- Batch sweep on long-range Copy at $b \in 8, 32, 256$

Depth scaling sweep – `complex_sigmoid` on LRA-ListOps L=1024 at $depth \in 2, 4, 6, 10, 14, 20$; H100, `batch=32`, `lr=1e-3` cosine over 30K steps, N=3 seeds. Detailed in §4.8 / §5.5.

Substrate ablations on PCT – A1 (native complex linear vs (`Re`, `Im`) two real linears), A2, A3, A4. Detailed in Appendix F via the public data archive.

Appendix 1. Data location

All raw results – per-seed metrics, configuration files, training logs, and aggregation scripts – for every test listed in Appendix 0 are available at:

<https://github.com/leohio/phase-coherent-transformer-r-d/tree/main/result>

The repository is organised by experiment family; a top-level `summary.md` gives a one-paragraph orientation per family.

Appendix 2. Contents of `result/summary.md`

This appendix reproduces the orientation document `result/summary.md` from the public data archive linked in Appendix 1. The full file is also available at <https://github.com/leohio/phase-coherent-transformer-r-d/blob/main/result/summary.md>.

Cross-task headline matrix

The headline finding for each task, with the best-performing cell shown. "PCT-fair" means $real_{dim} = complex_{dim} \times \sqrt{2} \approx 1.41$, screening cells with `softmax=OFF`, `batch=32`, N=3 seeds.

Task	Setup	Winner(s)	Score
Copy Memory d=1000 (small)	dim=128, N=3, PCT-fair	All cells saturate	1.00
Copy Memory d=500 (mid)	dim=256, N=3, PCT-fair	real_screen + complex_sigmoid (dual)	1.000 / 1.000
Copy Memory d=2000 (mid)	dim=256, N=3, PCT-fair	real_screen + complex_sigmoid (dual)	1.000 / 1.000
Copy Memory d=2000–5000	dim=256, batch=8, ≥ 17 seeds	complex_sigmoid uniquely	1.000
NIAH L=2048 (mid)	dim=256/L6, chunked attention, N=3	complex_sigmoid (PCT)	1.000
LRA-ListOps medium (small)	dim=64, N=3	complex_sigmoid	0.7188
LRA-ListOps L=1024	dim=128, N=3	complex_sigmoid	0.854
LRA-Text 4K	dim=128, N=3, PCT-fair	complex_sigmoid + complex_screen (dual)	1.000 / 1.000 (others ≤ 0.65)
LRA-Image CIFAR	dim=128, N=6, PCT-fair	complex_sigmoid	0.458
FFT-MNIST t=16	dim=128, N=3, PCT-fair	complex_screen + complex_sigmoid (dual)	0.938 / 0.927
Phase Memory	dim=64, N=3	real_screen + complex_sigmoid + complex_tanh+1 (tied)	1.000 / 0.995 / 1.000
Multi-Pitch synth	dim=128, N=3	complex_screen + complex_sigmoid	1.000 / 0.997
MusicNet real (small)	dim=64, N=3	complex_screen (marginal)	0.218
RadioML synth L1 / L2	dim=64, N=3	complex_sigmoid (L1) / complex_screen (L2)	0.598 / 0.637
RadioML real L1 / L2	dim=64, N=3	real_screen	0.363 / 0.389
Two-axis ($C2 \times C3$) isolation	dim=128, N=3	sigmoid 1.000 / softplus 1.000 / cubic 0.200 / clamped_relu 0.103 / ReLU 0.107	0.893 absolute gap (sigmoid $\rightarrow$ ReLU)
Depth scaling	dim=128, batch=32, N=3, $d \in \{2, 4, 6, 10, 14, 20\}$	complex_sigmoid trains at all depths	no depth-related accuracy collapse
Path-X	not yet run		–

Cross-task winner counts

How often each cell appears as winner / co-winner across the discriminating tasks above:

Cell	# times winner / co-winner	Notable strengths
complex_sigmoid (PCT)	9	Long-range Copy d=1000–5000, NIAH L=2048, all PCT-fair language/image tasks, LRA-ListOps medium
complex_screen	5	FFT-MNIST t=16, multi-pitch, LRA-Text 4K, LRA-ListOps medium close-2nd, harder-SNR RadioML synth
real_screen	4	Mid-scale Copy d=2000, real-RadioML at both levels, FFT-MNIST real-side leader
complex_softmax	0	
real_softmax	0	
real_sigmoid	0	

$\rightarrow$ **complex_sigmoid** + **complex_screen** + **real_screen** account for essentially all wins; vanilla softmax

and `real_sigmoid` never win discriminating tasks.

Top-line takeaways

1. **complex_sigmoid (PCT) is the most consistent single attention design across the tested task families** – content/phase retrieval, long-range memory, algorithmic, image, text, and audio (synth).
2. **complex_screen is competitive whenever the task structure rewards selectivity** – multi-source identification, phase-sensitive classification, language byte-token retrieval, harder-SNR modulation. On phase-sensitive tasks it ties or trails sigmoid by < 0.02 .
3. **real_screen is the only real-side cell that breaks out of chance / baseline on most tasks** – it is the strongest baseline for showing real-vs-complex gaps without confounding by activation choice, and is the unique winner on Real RadioML L1/L2.
4. **Vanilla softmax is structurally limited on long-range retrieval** – it cannot solve Copy $d=2000$ at any tested capacity. PCT deeply solves NIAH $L=2048$ – i.e., a complex-disadvantaged purely positional task is solved by a complex-attention design.
5. **ReLU attention is catastrophic on phase / long-range** – anti-correlation information is destroyed by the dead negative half.
6. **The complex substrate is necessary existentially, not in any specific component** – at least one of Q, K, V, embed/output must be complex-valued; the four-component substrate ablation shows no single component carries the substrate irreducibly.
7. **The framework requires four conditions, all read in operating-range form:** (C1) real-valued gate output, (C2) bounded gate on the operating range, (C3) gradient nonzero on the operating range, (C4) element-independent gate. The 2×2 isolation experiment confirms all four are independently necessary; **C3 is the dominant failure axis**, but **C2 is also a real condition**, fatal at large M.
8. **PCT does not exhibit a depth-related accuracy collapse** across depths $d \in 2, 4, 6, 10, 14, 20$ on LRA-ListOps $L=1024$ ($10\times$ param range, $0.40\text{ M} \rightarrow 3.96\text{ M}$). This is a *qualitative* claim, not a quantitative power-law scaling claim.

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Complex-valued neural networks

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